

WHY E-WASTE Cannot Be Taken Lightly



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Electronic waste (e-waste) or waste electrical and electronic equipment (WEEE) is a term used to cover all electrical or electronic equipment and their parts that have been discarded by their users as waste without the intent of re-use. E-waste includes a wide range of products; almost any household or business item with circuitry or electrical components with power or battery supply can form e-waste.

E-waste contains various materials including hazardous, valuable and scarce metals. Common hazardous materials found in e-waste are heavy metals such as lead, mercury and cadmium and chemicals like ozone-depleting CFC or various flame retardants.

A gold mine

In addition, e-waste contains many valuable materials such as copper, iron, aluminium and plastics, and precious metals like gold, silver, palladium and platinum, which can be recycled. In fact, up to 60 elements from the periodic table can be found in complex electronic products, and many of these are recoverable. So from a resource perspective, e-waste is a potential urban mine that could provide a great amount of secondary resources for manufacture, refurbishment and recycling.

URBAN MINE: AN ESTIMATE OF MATERIAL VALUE OF E-WASTE IN 2014

Material	Quantity (kilotonnes)	Price (million Euros)
Precious metals:		
Gold (Ag)	0.3 (300 tonnes)	10,400
Palladium (Pd)	0.1 (100 tonnes)	1800
Silver (Ag)	1	580
Metals:		
Copper (Cu)	1900	10,600
Iron, Steel (Fe)	16,500	9000
Aluminium (Al)	220	3200
Plastics:		
PP, ABS, PC, PS and others	8600	12,300
<i>Estimated total price: € 48 billion</i>		

For instance, according to United Nations University (UNU), a worldwide leading institute in e-waste research, gold content in the estimated e-waste generated in 2014 is roughly 300 tonnes, which is approximately 11 per cent of the global gold production from mines in 2013 (2770 tonnes, according to USGS 2014).

According to UNU-IAS, it is estimated that the total amount of e-waste generated globally in 2014 was 41.8 million metric tonnes (MT); India's share being 1.7MT. The amount of e-waste has been forecast to increase to 50MT by 2018. A report by United Nations Environment Program (UNEP) mentions that the amount of e-waste being produced from goods like mobile phones, laptops and computers could go up by 500 per cent over the next decade in some countries including India.

The intrinsic material value of global e-waste generation in 2014 alone was estimated at € 48 billion. Material value is dominated by metals like gold, copper, iron, silver and palladium and plastic contents. Refer table above for a detailed estimate.

The danger

Harvesting e-waste can be dangerous if it is not recycled using state-of-the-art technology.

Toxic e-waste dumped in a landfill

